Aliaksandr (Alex) Melnichenka

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Education

Berea College — **May 2028**
B.S. in Physics and Mathematics (double major), minor in Computer Science. Dean's List every term. — Berea, KY

Lyceum of Belarusian State University — **2022 – 2024**
STEM magnet high school, physics track. — Minsk, Belarus

Publications and Preprints

[1] **A. Melnichenka***, P. Liong*, A. Bukhtatyi, A. Bilous, L. Levitov. "Spontaneous Running Waves and Self-Oscillatory Transport in Dirac Fluids." *Under review at Physical Review Letters*, arXiv:2512.16571, 2025. *Equal contribution.

[2] **A. Melnichenka**, A. Lazarian, D. Pogosyan. "Recovering 3D Magnetic Turbulence from a Single Polarization Map." *Submitted to The Astrophysical Journal*, arXiv:2602.22204, 2026.

Research Experience

Electron-Fluid Instabilities in Dirac Materials — MIT, Condensed Matter Theory
Undergraduate Researcher with Prof. Leonid Levitov — *2025 – present*

First author on a manuscript under review at *Physical Review Letters* on current-driven self-sustained waves and self-oscillatory transport in Dirac electron fluids.

Developed the full computational pipeline: pseudospectral PDE solver, collision-operator eigenmode computations on large grids, instability-threshold mapping, and wavelength-selection analysis across multiple initial conditions.

Produced the simulations and core numerical evidence for the project; work was presented at the 2026 APS Global Physics Summit by Prof. Leonid Levitov.

Turbulence Recovery from Radio Polarization — UW-Madison, Astronomy
Undergraduate Researcher with Prof. Alex Lazarian and Prof. Dmitri Pogosyan — *2025 – present*

Developed a single-frequency polarization diagnostic that recovers 3D inertial-range magnetic turbulence statistics from one radio map. First-author manuscript submitted to *The Astrophysical Journal*.

Validated on synthetic Faraday screens and AthenaK MHD turbulence simulations in sub-Alfvénic and super-Alfvénic regimes.

Built a reproducible pipeline with checks for interferometric filtering, ready for LOFAR, MeerKAT, and VLA data.

Self-Interacting Dark Matter near Supermassive Black Holes — MIT Kavli Institute for Astrophysics
Undergraduate Researcher with Prof. Mark Vogelsberger (mentor: Xuejian Shen) — *2025 – present*

Reproduced published benchmarks for SIDM evolution near supermassive black holes and isolated discrepancies via targeted ablation tests. Developing an orbit-averaged stochastic loss-cone model with scaling predictions for collapse timescales.

Selected Talks

American Astronomical Society, 247th Meeting — Phoenix, AZ
Oral presentation on single-frequency Faraday-screen tomography — *Jan 2026*

American Physical Society, Division of Plasma Physics — Long Beach, CA
Contributed oral talk on 3D magnetic turbulence recovery from polarization maps — *Nov 2025*

The Magnetized Turbulent Universe (Conference Honoring A. Lazarian) — Playa del Carmen, Mexico
Invited talk on polarization-angle statistics and crossover scaling. Only undergraduate speaker. — *Nov 2025*

Honors, Fellowships, and Selected Programs

Emergent Ventures Fellowship (23,000$), Mercatus Center at George Mason University, selected by Tyler Cowen, 2026.
Sigma Pi Sigma, national physics honor society; first sophomore in department's 96-year history inducted, 2026.
Featured by LEX 18 News for building free science education platforms used by students internationally, 2026.
Caltech SURF Research Fellow, selected to work with Prof. Jim Fuller, 2026.
International Theoretical Physics Practicum, returning participant, 2026.
Putnam Mathematical Competition, score 28, top 13% nationally, 2025.
Belarus National Physics Olympiad: Gold medal (2022), Silver medal (2024), Bronze medal (2023).
IPhO Reserve Training Camp, top six nationally in Belarus, 2022.
First Degree Diploma, Phystech.International (Moscow Institute of Physics and Technology), 2023.
Presidential Award for Gifted Youth (Belarus), 2022.

Projects, Leadership, and Outreach

SavchenkoSolutions.com Founder

Collaborative solution platform for Savchenko's Problems in Physics *2023 – present*

Built a community-maintained platform around a canonical advanced problem collection: 67 contributors from 40 countries, 1,493 published solutions, and roughly 1,500 daily users.

Personally authored 847 solutions and produced the first English translations for 362 problems; designed the editorial workflow, review queue, and bilingual publication standards.

BelSO.org Founder and Lead Engineer

National problem archive and training platform for Belarusian science olympiads *2023 – present*

Built a unified platform spanning physics, mathematics, astronomy, and informatics, with 8,500 problems, 3,200 solutions, and roughly 12,500 monthly users.

Used by olympiad students and coaches across Belarus; recognized on International Physics Olympiad country resource pages.

Software and Technical Skills

Languages and Tools Python (NumPy, SciPy, Astropy, JAX), JavaScript, HTML/CSS, C, Bash, Git, LaTeX

Methods Spectral PDE solvers, turbulence statistics, Monte Carlo and Langevin methods, full-stack web development, polarization correlation analysis, structure functions, numerical ODEs and PDEs

Open-Source Packages AstroTurbulence (polarization statistics), electronic-kapitsa-waves (spectral synthesis), SIDM_Transport_Theory_vs_MC (drift diffusion benchmarks). All with commit-pinned figures and environment specifications.

Human Languages English (fluent), Belarusian (native), Russian (native)

Teaching

Berea College, Physics Department Berea, KY

Teaching Assistant *Jan 2025 – present*

Led weekly office hours and review sessions for introductory and calculus-based physics. Coached students on scientific writing, error checking, and reproducible computational workflows.